

SEAT No. _____

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[56]

SARDAR PATEL UNIVERSITY
BCA Semester-5 Examination, 2020
US05CBCA23 || SOFTWARE ENGINEERING
26th December, 2020.

Time : 02.00 p.m. – 04.00 p.m.

Marks: 70

Q-1 Select the appropriate option for the following questions:

[10]

- 1) Which phase is required to understand the problem?
A. System Design C. Requirement Analysis
B. Coding D. Testing
- 2) _____ model provides better risk management and cost of each phase.
A. Spiral C. Prototype
B. Iterative enhancement D. Waterfall
- 3) _____ activity is used to understand the needs, goals and constraints.
A. Testing C. Design
B. Requirement Specification D. Problem Analysis
- 4) _____ characteristic of SRS means the entire requirement denotes one interpretation.
A. Complete C. Unambiguous
B. Reliability D. Traceable
- 5) Which is not factor for design specification?
A. Problem Specification C. Abstraction
B. Design Decision D. Module Specification
- 6) Which is not the level of Cohesion?
A. Logical C. Physical
B. Sequential D. Coincidental
- 7) In structured design methodology the hierarchy of modules is represented by the _____.
A. flow chart C. Gant chart
B. PERT chart D. structure chart
- 8) The information hiding principle in modern programming languages by _____.
A. data-hiding C. encapsulation
B. data-abstraction D. inheritance
- 9) _____ is the collection of computer programs, procedures and data.
A. Hardware C. Network
B. Software D. Engineering
- 10) An SRS provides a reference for _____ of the final product.
A. Validation C. Verification
B. Xerox copy D. Quality

Q-2 Tick (TRUE or FALSE) and Fill in the blanks.

[08]

- 1) Prototype model reduce the first two limitations of Waterfall model. [TRUE/FALSE]
- 2) The coding should follow rules of _____.

[PTO]

[1]

- 3) COCOMO stands for: _____
- 4) An SRS establishes the basis for agreement between the _____ and the _____.
- 5) Structured design methodology tries to reduce coupling. [TRUE/FALSE]
- 6) Number of subordinates associated with given module is known as _____.
- 7) A failure is produced only when there is a fault in the system. [TRUE/FALSE]
- 8) Structured programming is often called _____ programming.

Q-3 Give short answers of the following questions: (ANY TEN)

[20]

- 1) Explain advantages of spiral model.
- 2) List down characteristics of software process.
- 3) Define: Software, Software Engineering.
- 4) What is Structured English?
- 5) List the components of SRS.
- 6) "A high quality SRS reduces development cost." Justify.
- 7) List the levels of Cohesion.
- 8) Define: Coupling and Cohesion.
- 9) Differentiate between Top-down and Bottom-up approaches.
- 10) Write the goal of coding.
- 11) What do you mean by information hiding?
- 12) List any four automated testing tools.

Q-4 Give Long answers of the following questions: (ANY FOUR)

[32]

- 1) Explain waterfall model with advantages and limitations.
- 2) Write short notes on design and testing phase of software development.
- 3) What is SRS? Explain needs and general characteristics of SRS.
- 4) Explain COCOMO Model.
- 5) Write a short note on Cohesion.
- 6) Write a short note on Coupling.
- 7) Explain coding approach and programming style.
- 8) What is testing? Explain levels of testing using diagram.

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[2]